

Iris Dataset

Learnings:

- hue is for compairing/analysing using third parameter.
- we can add different color to bars or a graph using palette
- we can use desired colors, just make a variable containing color names and assign this to a palette.
- for line concatenation using \ ,no comment or anything not even space should be allowed after "
- use ; to get clean output figure
- when we make some change in dataframe like using drop() to remove column use argument inplace if True replaces the original file with modified one(default-False)
- df.plot.area from pandas, works only for numerical data

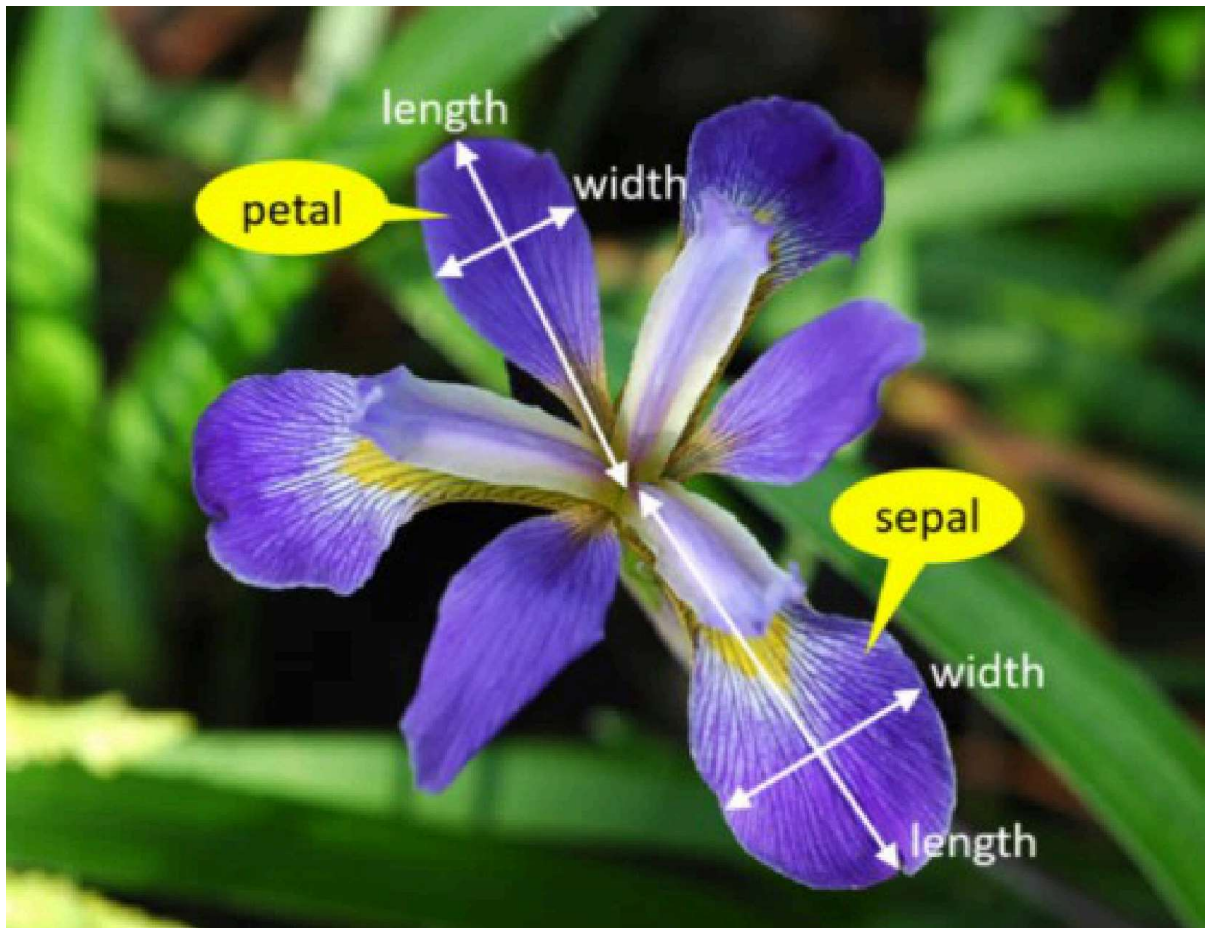
```
In [4]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

import warnings
warnings.filterwarnings('ignore')           # it won't display warning messages in n
```

```
In [9]: import PIL.Image    # library to access image
```

```
In [23]: iris_flower_image = PIL.Image.open(r"C:\Users\Avinash\Downloads\Iris flower_page-00
iris_flower_image
```

Out[23]:



```
In [15]: iris = pd.read_csv(r"C:\Users\Avinash\OneDrive\Desktop\Vaishnavi_FSDS\downloads\Iri
```

```
In [17]: iris
```

Out[17]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...	...
145	146	6.7	3.0	5.2	2.3	Iris-virginica
146	147	6.3	2.5	5.0	1.9	Iris-virginica
147	148	6.5	3.0	5.2	2.0	Iris-virginica
148	149	6.2	3.4	5.4	2.3	Iris-virginica
149	150	5.9	3.0	5.1	1.8	Iris-virginica

150 rows × 6 columns

In [19]: `iris.info()` *# quick summery*

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Id               150 non-null   int64
1   SepalLengthCm    150 non-null   float64
2   SepalWidthCm     150 non-null   float64
3   PetalLengthCm    150 non-null   float64
4   PetalWidthCm     150 non-null   float64
5   Species          150 non-null   object
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB
```

In [25]: `iris.drop('Id',axis=1,inplace=True)` *# inplace(True)- modification made in*
`iris.head()`

Out[25]:

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa

```
In [27]: iris1 = pd.read_csv(r"C:\Users\Avinash\OneDrive\Desktop\Vaishnavi_FSDS\downloads\Ir
iris1

# original file do not change due to drop(' ', axis= , inplace=True)
```

```
Out[27]:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...	...
145	146	6.7	3.0	5.2	2.3	Iris-virginica
146	147	6.3	2.5	5.0	1.9	Iris-virginica
147	148	6.5	3.0	5.2	2.0	Iris-virginica
148	149	6.2	3.4	5.4	2.3	Iris-virginica
149	150	5.9	3.0	5.1	1.8	Iris-virginica

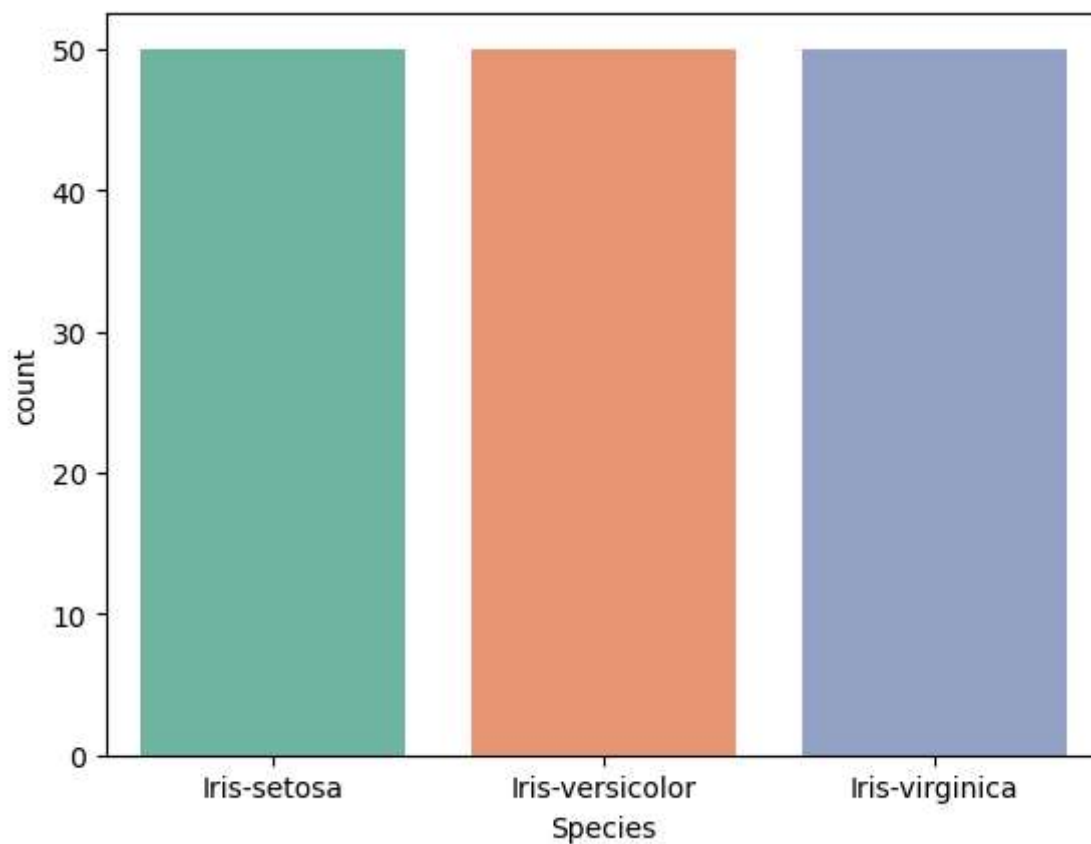
150 rows × 6 columns

```
In [31]: iris['Species'].value_counts() # count unique value: series of va
```

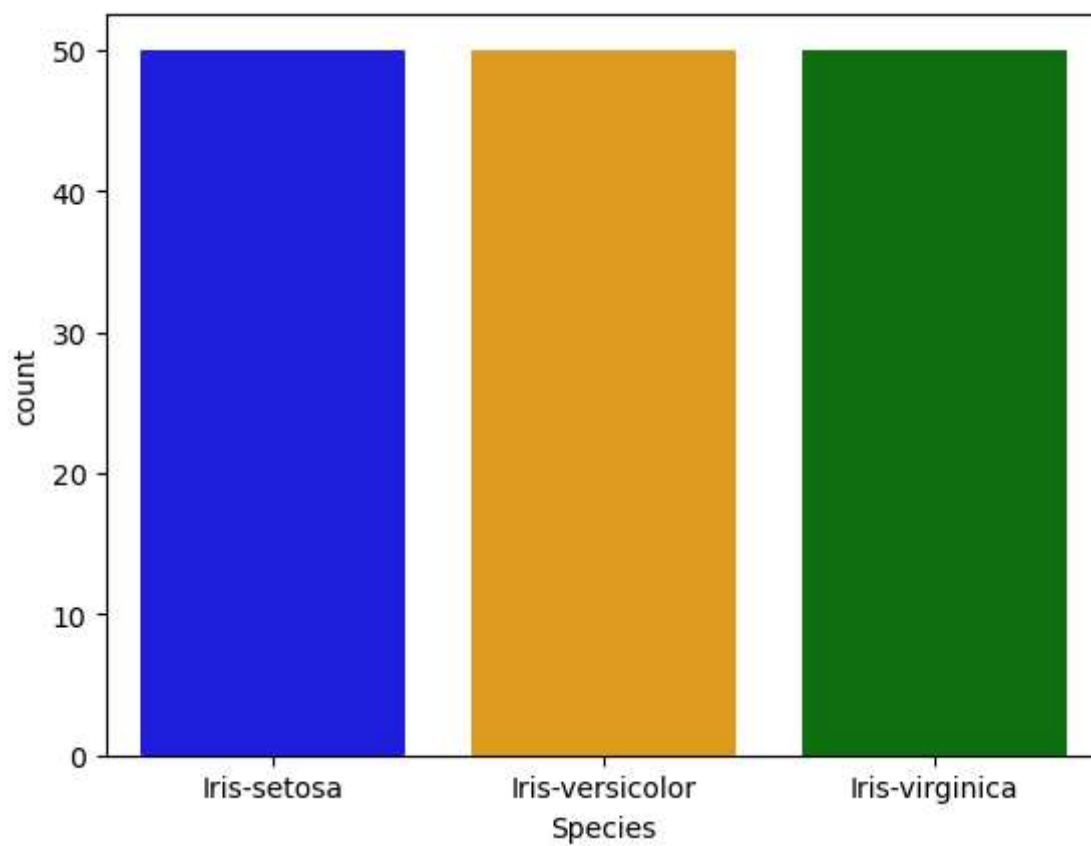
```
Out[31]: Species
Iris-setosa      50
Iris-versicolor  50
Iris-virginica   50
Name: count, dtype: int64
```

```
In [ ]: sns.countplot(data=iris,x='Species', palette='Set2') # height represents coun
plt.show()
```

```
In [53]: sns.countplot(data=iris,x='Species', palette='Set2') # to get different color for
plt.show()
```



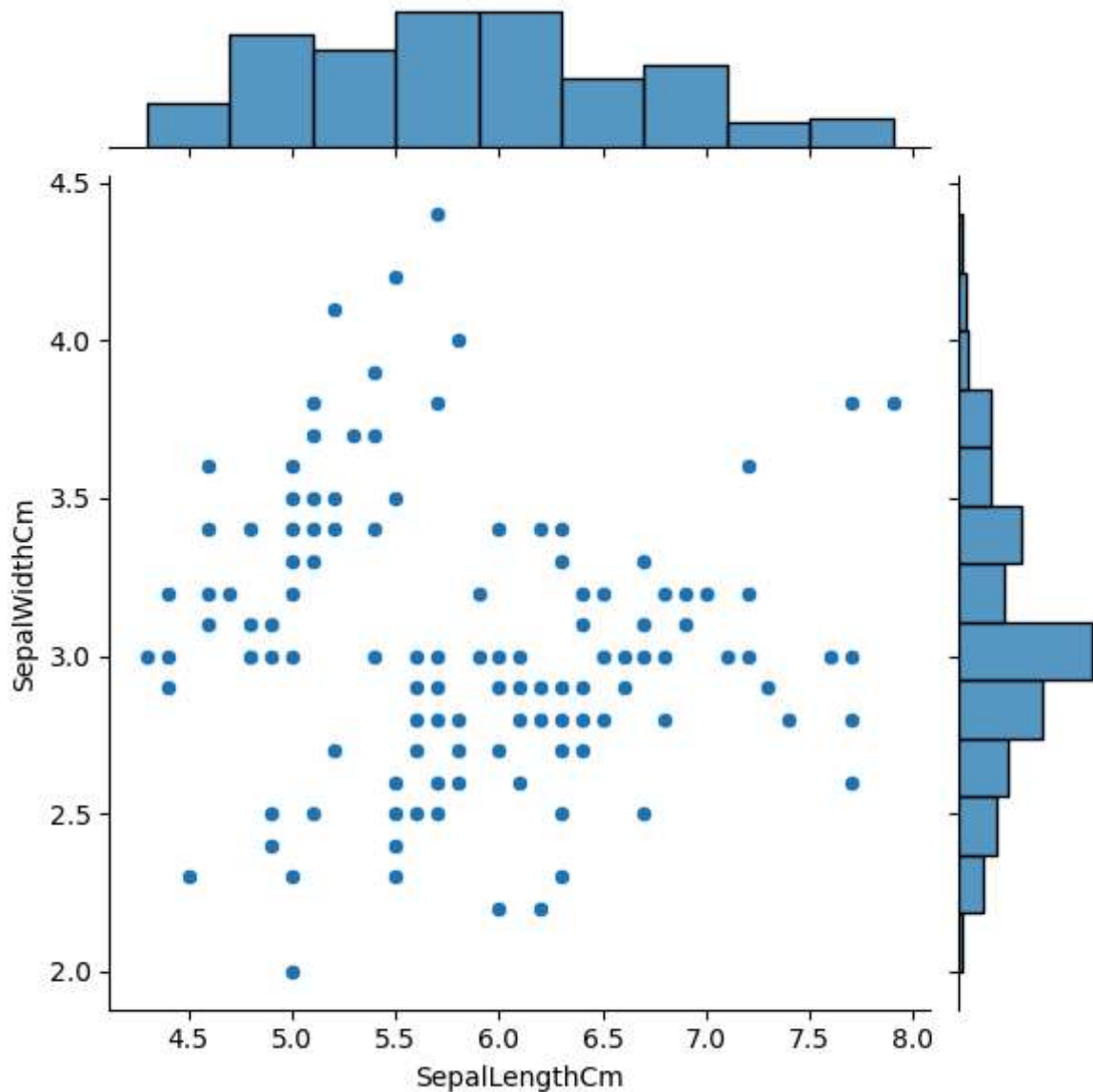
```
In [65]: color = ('blue', 'orange', 'green') # to get different desire  
sns.countplot(data=iris, x='Species', palette=color) # height represents count  
plt.show()
```



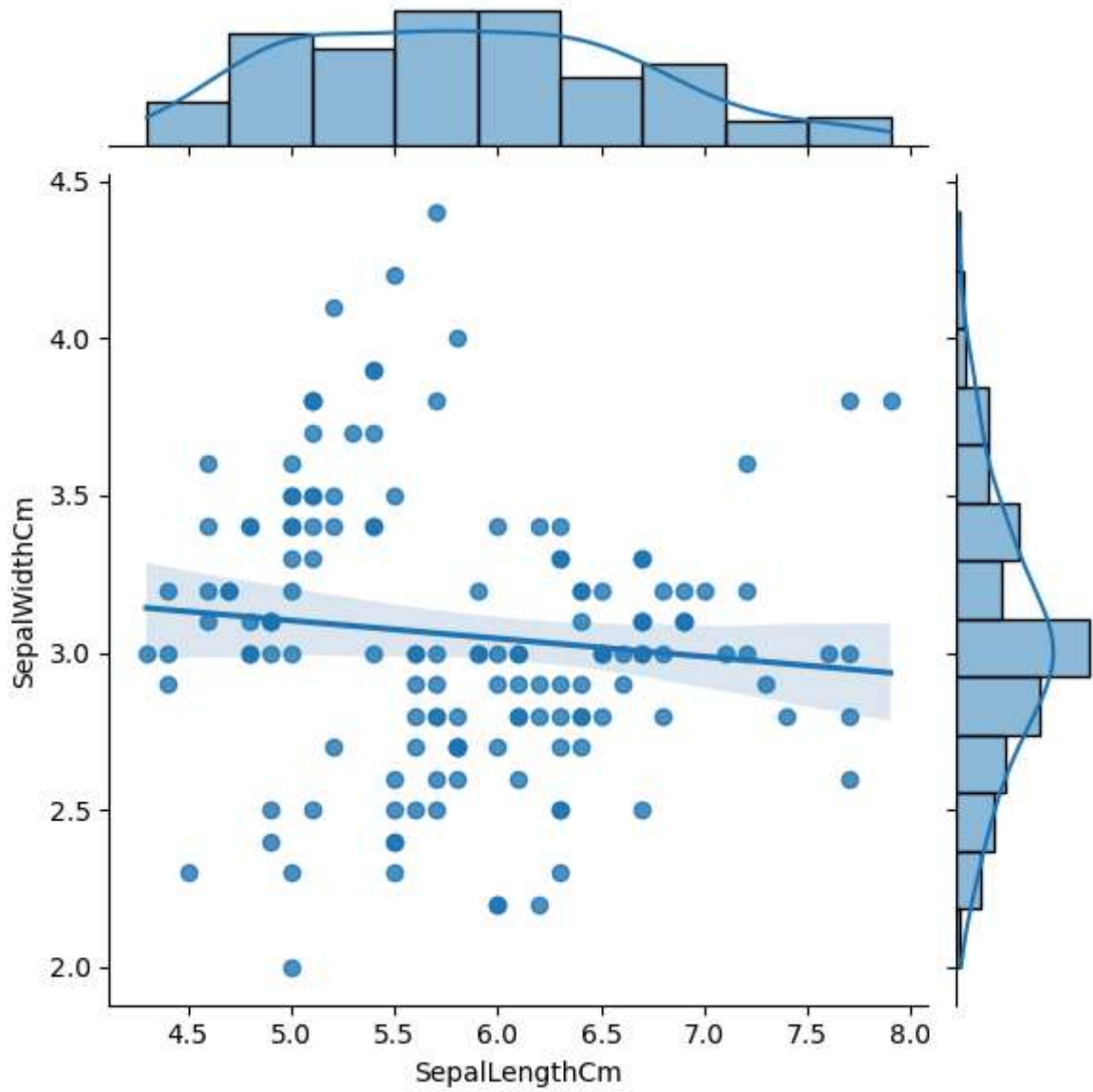
```
In [55]: iris.columns
```

```
Out[55]: Index(['SepalLengthCm', 'SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm',  
              'Species'],  
             dtype='object')
```

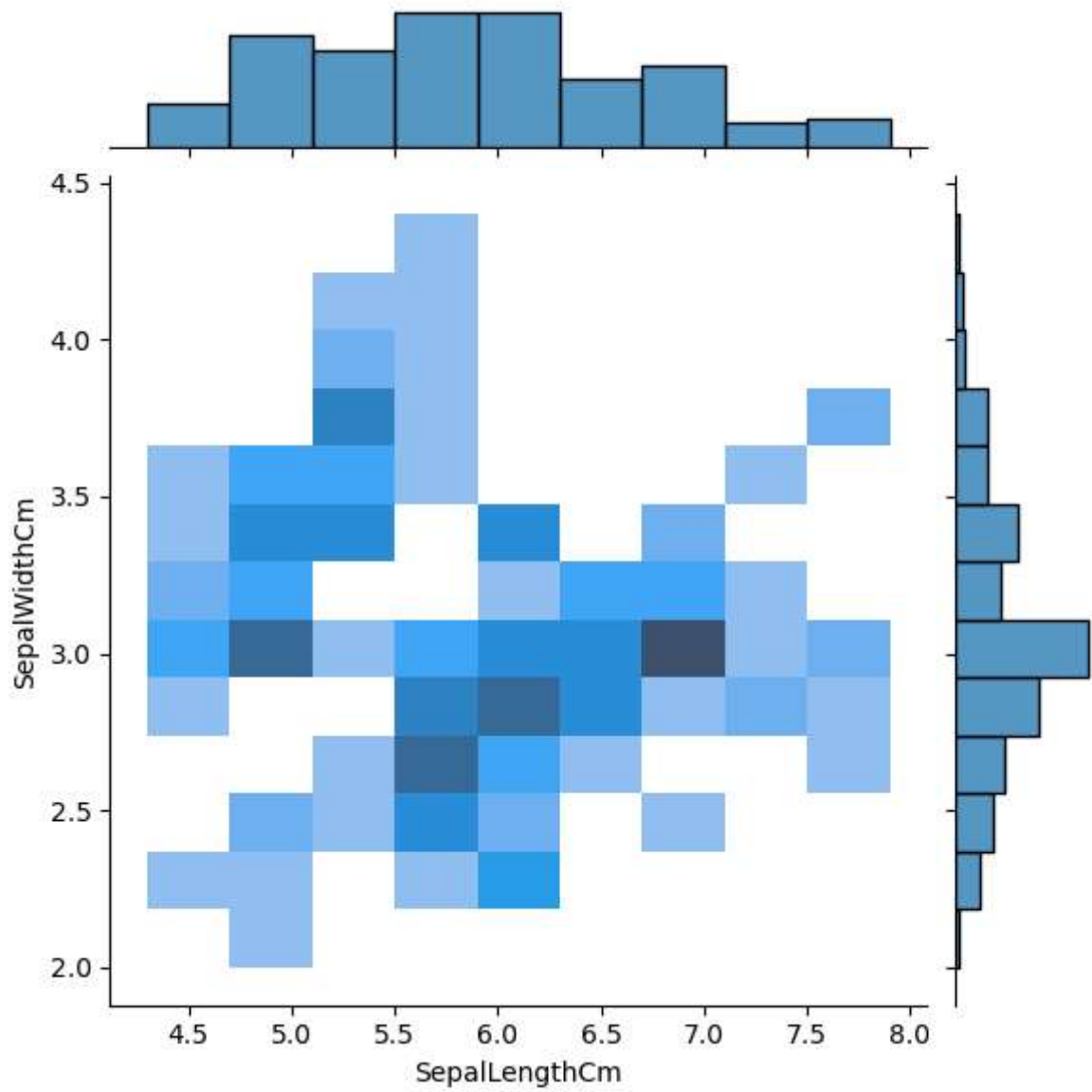
```
In [57]: fig = sns.jointplot(x='SepalLengthCm',y='SepalWidthCm',data=iris)
```



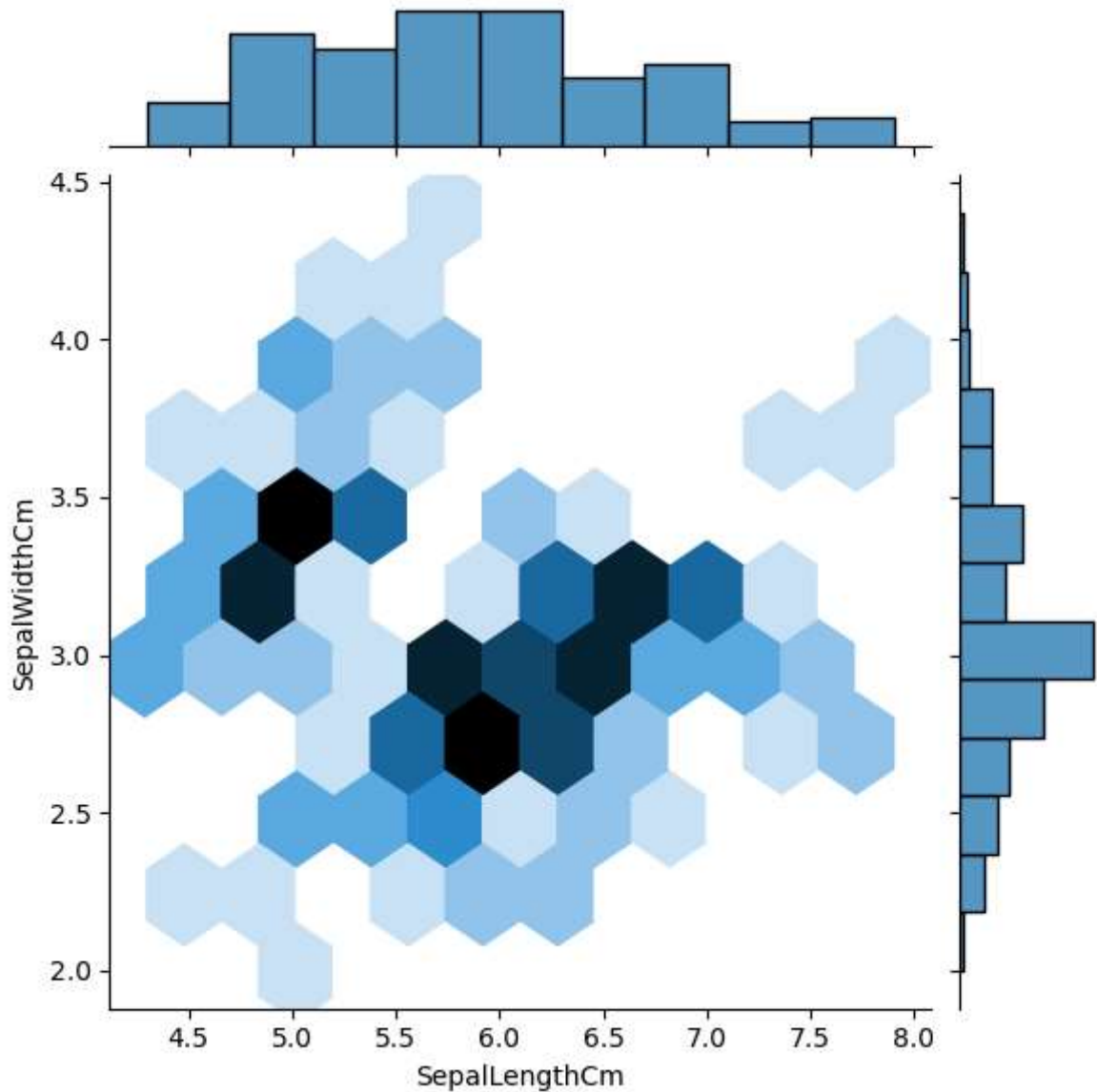
```
In [61]: fig = sns.jointplot(x='SepalLengthCm',y='SepalWidthCm',data=iris,kind='reg')
```



```
In [67]: fig = sns.jointplot(x='SepalLengthCm',y='SepalWidthCm',data=iris,kind='hist')
```



```
In [69]: fig = sns.jointplot(x='SepalLengthCm',y='SepalWidthCm',data=iris,kind='hex')
```

```
In [75]: import matplotlib.pyplot as plt
%matplotlib inline

sns.FacetGrid(iris,hue='Species',size=5)\
.map(plt.scatter,'SepalLengthCm','SepalWidthCm')\
.add_legend()
```

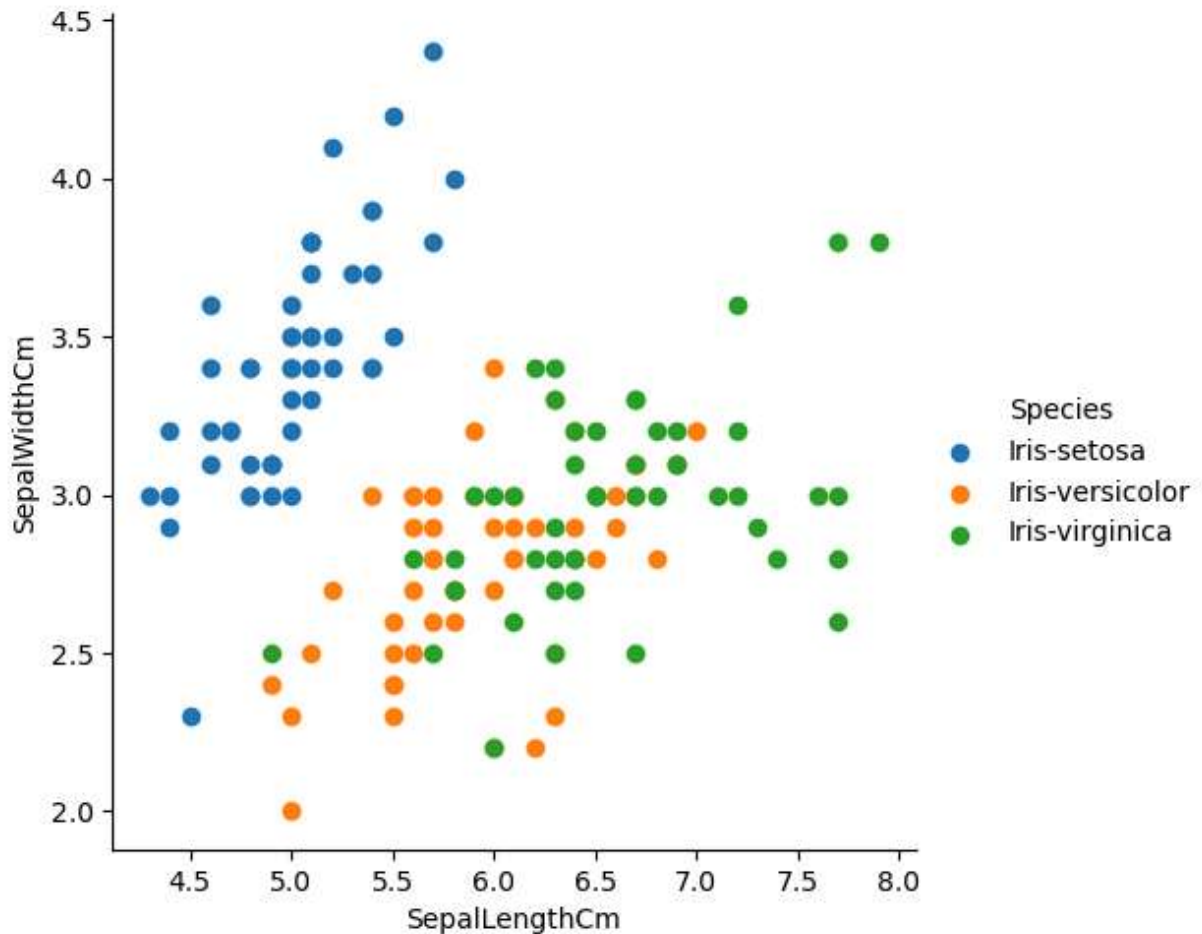
```
-----
TypeError                                Traceback (most recent call last)
Cell In[75], line 4
      1 import matplotlib.pyplot as plt
      2 get_ipython().run_line_magic('matplotlib', 'inline')
----> 4 sns.FacetGrid(iris,hue='Species',size=5)\
      5 .map(plt.scatter,'SepalLengthCm','SepalWidthCm')\
      6 .add_legend()

TypeError: FacetGrid.__init__() got an unexpected keyword argument 'size'
```

```
In [77]: sns.__version__           # check version
```

Out[77]: '0.13.2'

```
In [93]: sns.FacetGrid(iris,hue='Species',height=5)\
        .map(plt.scatter,'SepalLengthCm','SepalWidthCm')\
        .add_legend()\
        plt.show()\
        # in new version (0.9.0 and later) size replaced by height
```

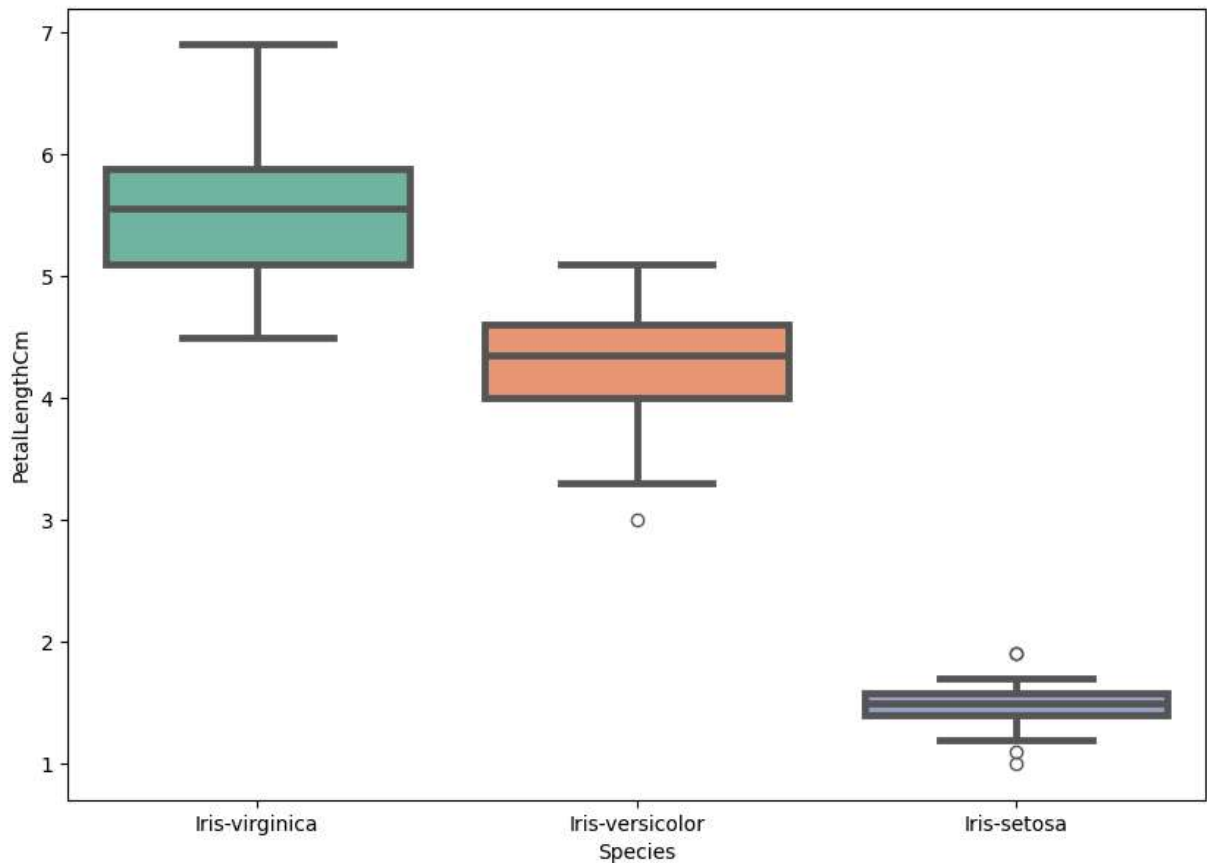


```
In [95]: iris.head()
```

```
Out[95]:
```

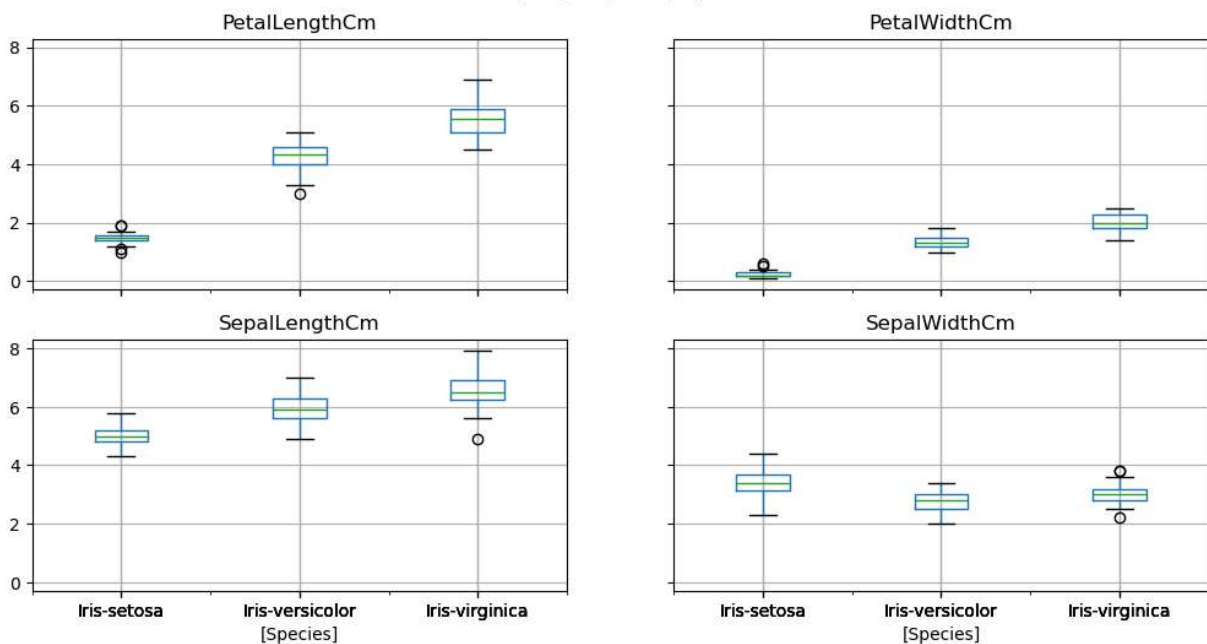
	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa

```
In [113... fig = plt.gcf()
fig.set_size_inches(10,7)
fig = sns.boxplot(x='Species',y='PetalLengthCm',data=iris,order=['Iris-virginica',
plt.show()
```

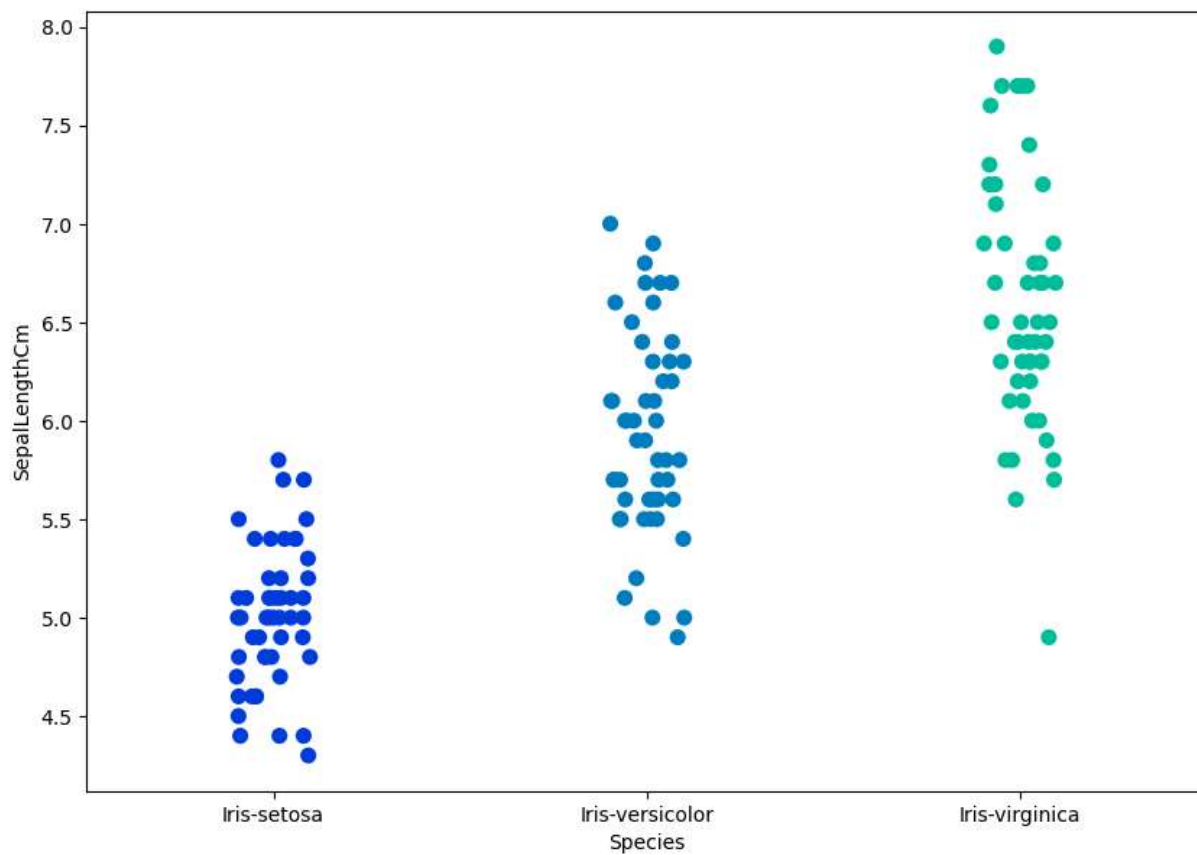


```
In [121... # iris.drop('Id',axis=1).boxplot(by='Species',figsize=(12,6))
iris.boxplot(by='Species',figsize=(12,6)) # it is not sns plotting
plt.show()
```

Boxplot grouped by Species



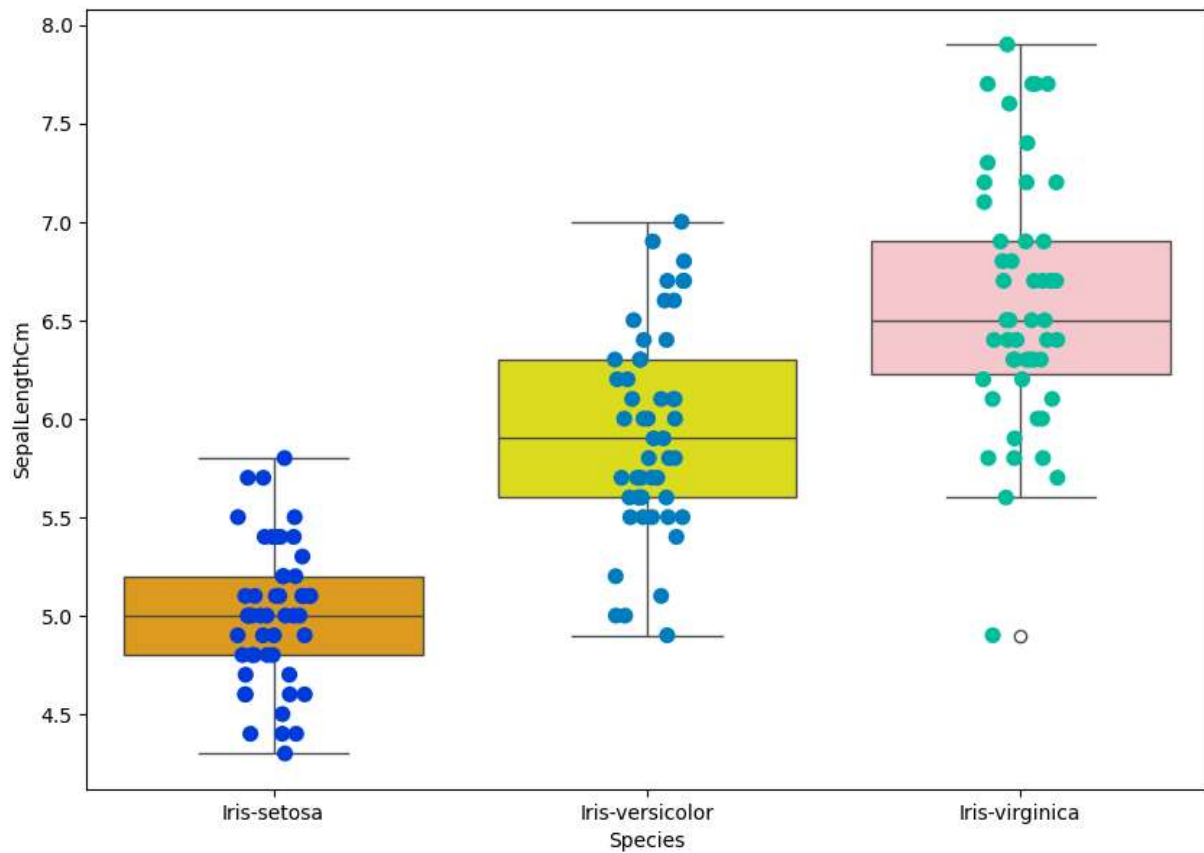
```
In [145... fig = plt.gcf()
fig.set_size_inches(10,7)
fig = sns.stripplot(x='Species', y='SepalLengthCm', data=iris, jitter = True, edgec
plt.show()
```



In [167...

```
fig = plt.gcf()
fig.set_size_inches(10,7)

color=('orange', 'yellow','pink')
fig = sns.boxplot(x='Species',y='SepalLengthCm',data=iris,palette=color)
fig = sns.stripplot(x='Species', y='SepalLengthCm', data=iris, jitter = True, edgec
plt.show()
```



In [173...

```

ax = sns.boxplot(x='Species',y='PetalLengthCm',data=iris)
fig = sns.stripplot(x='Species', y='SepalLengthCm', data=iris, jitter = True, edgec

boxtwo = ax.artists[1]
boxtwo.set_facecolor('yellow')
boxtwo.set_edgecolor('black')
boxthree = ax.artists[2]
boxthree.set_facecolor('red')
boxthree.set_edgecolor('black')
boxthree = ax.artists[0]
boxthree.set_facecolor('green')
boxthree.set_edgecolor('black')

plt.show()

```

```

-----
IndexError                                Traceback (most recent call last)
Cell In[173], line 4
      1 ax = sns.boxplot(x='Species',y='PetalLengthCm',data=iris)
      2 fig = sns.stripplot(x='Species', y='SepalLengthCm', data=iris, jitter = True,
    e, edgecolor='grey', size=8, palette='winter',orient='v')
----> 4 boxtwo = ax.artists[1]
      5 boxtwo.set_facecolor('yellow')
      6 boxtwo.set_edgecolor('black')

File ~\anaconda3\Lib\site-packages\matplotlib\axes\_base.py:1453, in _AxesBase.ArtistList.__getitem__(self, key)
    1452 def __getitem__(self, key):
-> 1453     return [artist
    1454             for artist in self._axes._children
    1455             if self._type_check(artist)][key]

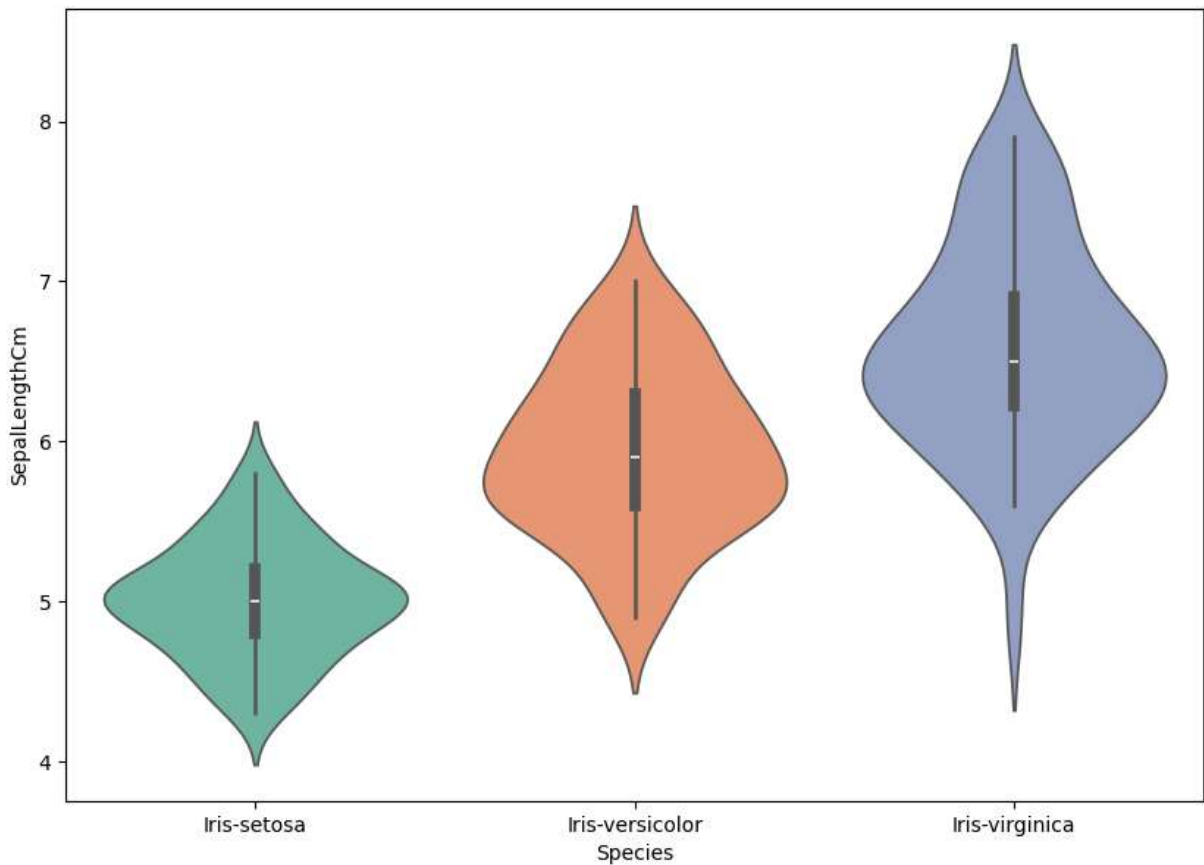
IndexError: list index out of range

```

```

In [181... fig = plt.gcf()
fig.set_size_inches(10,7)
fig1 = sns.violinplot(x='Species',y='SepalLengthCm',data=iris,palette='Set2')
plt.show()

```



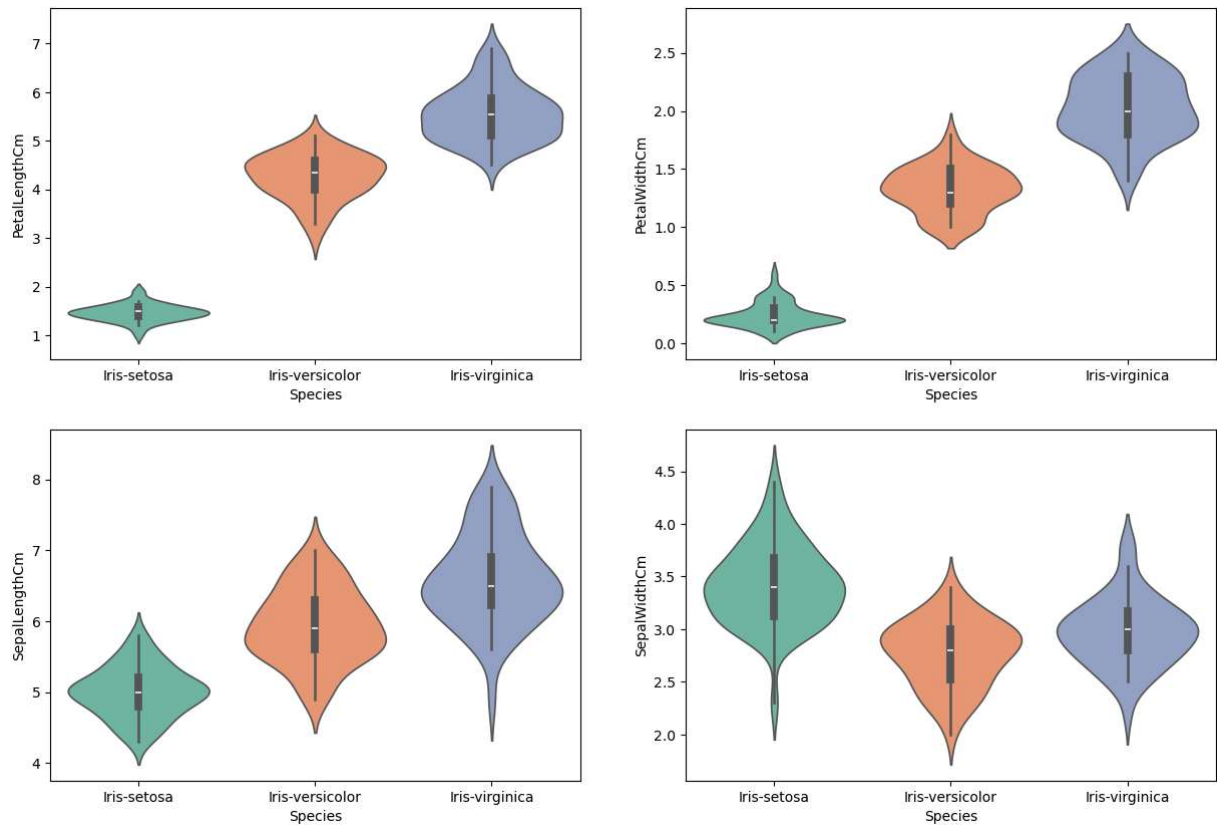
```

In [187... plt.figure(figsize=(15,10))
plt.subplot(2, 2, 1)                                # not a subplots
sns.violinplot(x='Species', y='PetalLengthCm', data=iris, palette='Set2')
plt.subplot(2, 2, 2)                                # not a subplots
sns.violinplot(x='Species', y='PetalWidthCm', data=iris, palette='Set2')

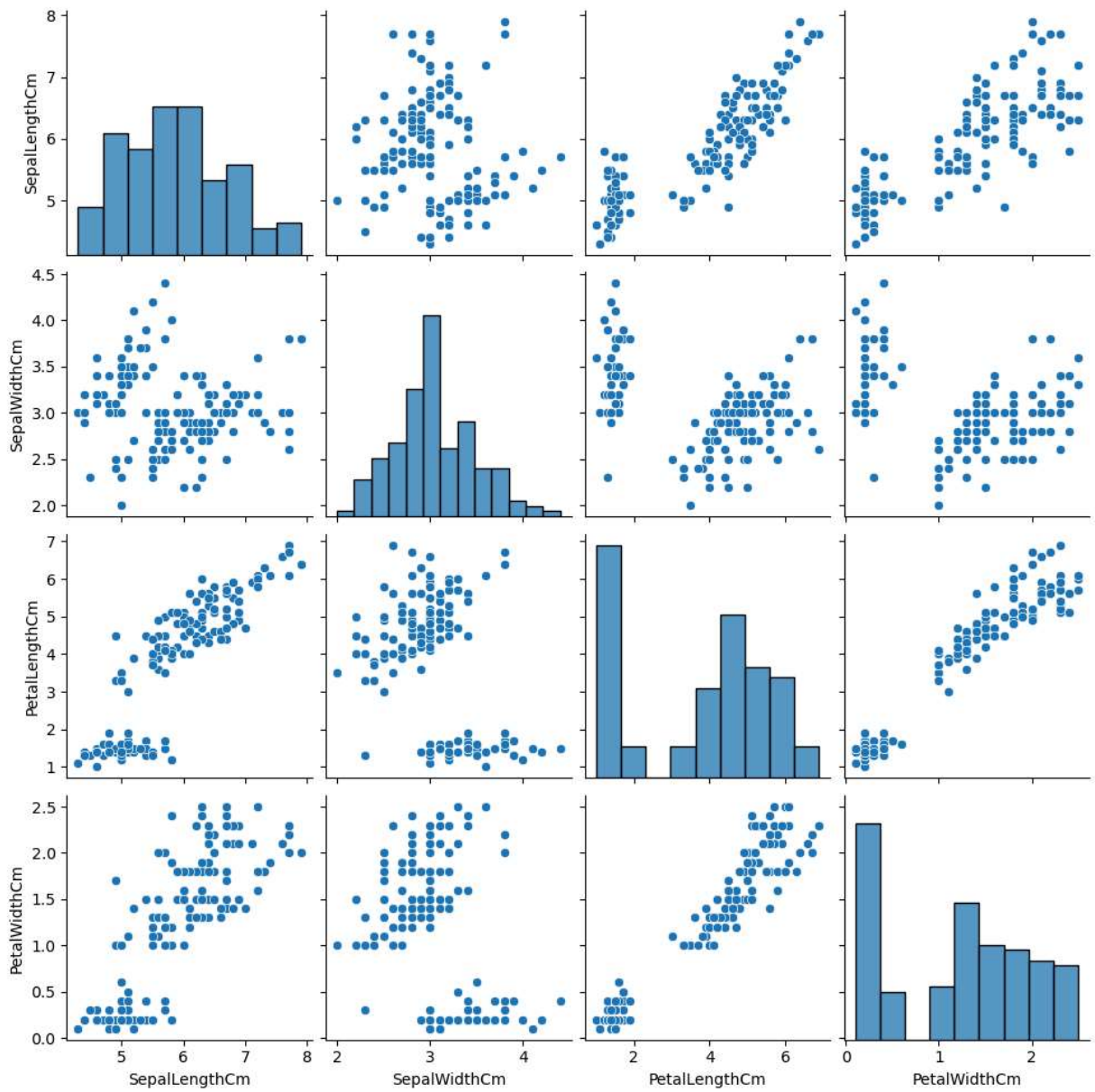
```

```
plt.subplot(2, 2, 3)                                # not a subplots
sns.violinplot(x='Species', y='SepalLengthCm', data=iris, palette='Set2')
plt.subplot(2, 2, 4)                                # not a subplots
sns.violinplot(x='Species', y='SepalWidthCm', data=iris, palette='Set2')

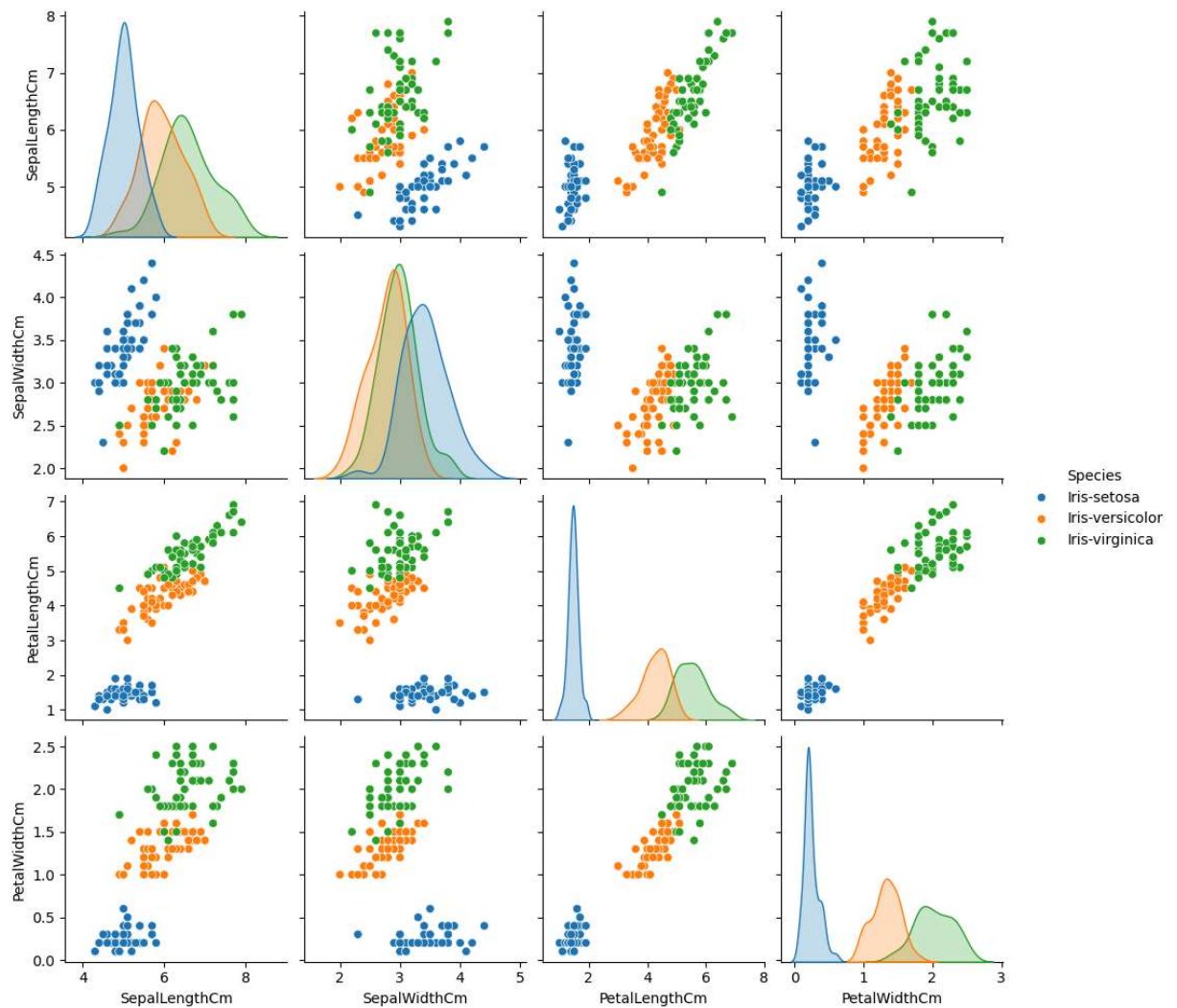
plt.show()
```



In [195... `sns.pairplot(data=iris, kind='scatter')`
`plt.show()`



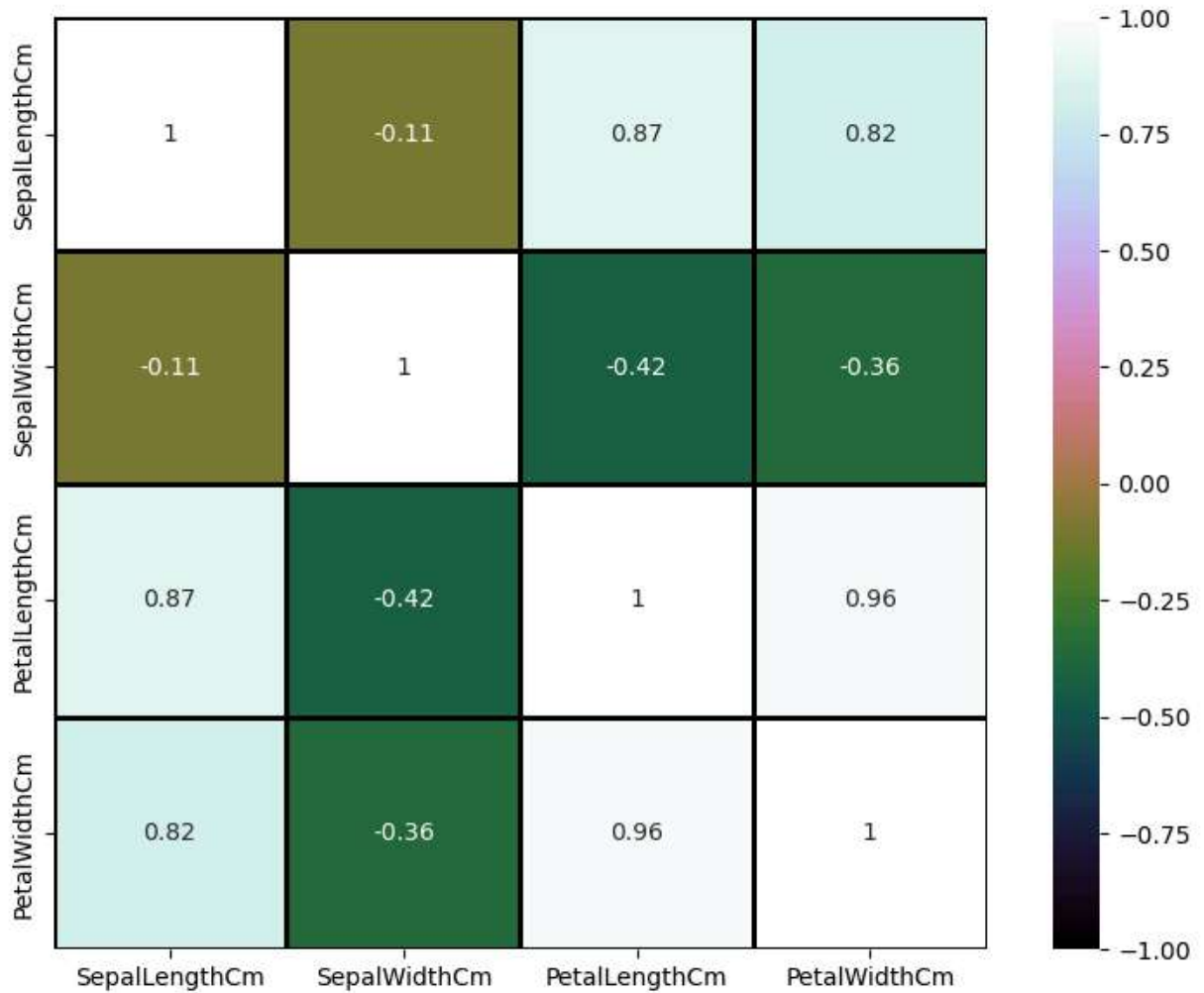
```
In [203... sns.pairplot(iris,hue='Species')  
plt.show()
```

In [229...

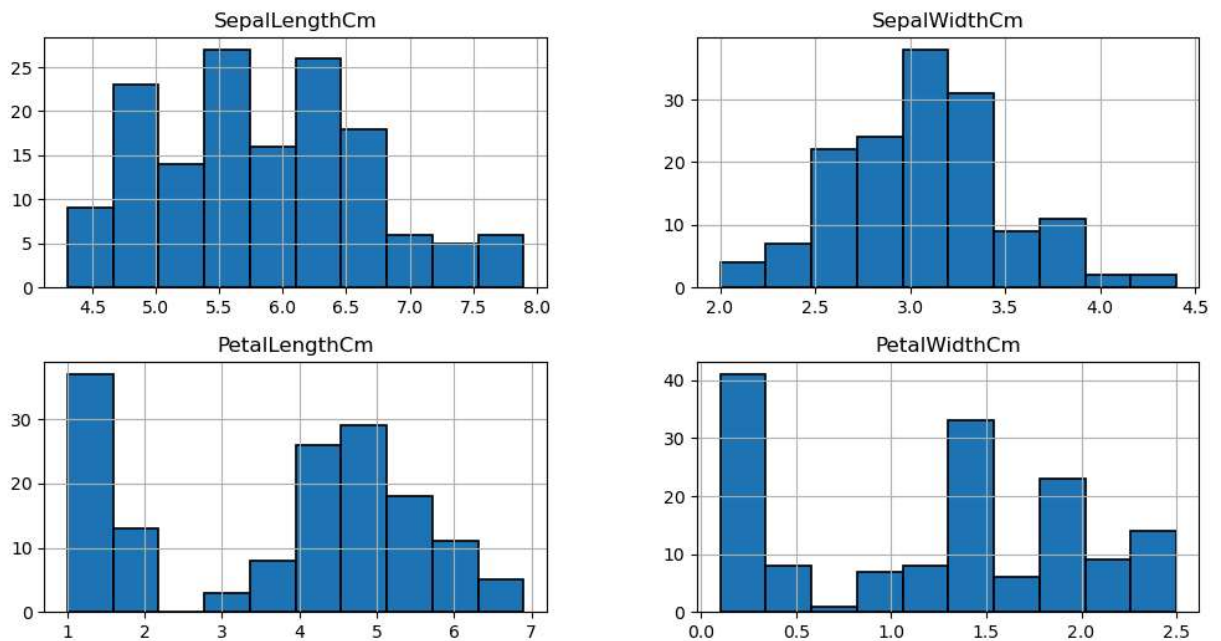
```
fig=plt.gcf()
fig.set_size_inches(10,7)
iris1= iris.drop('Species',axis=1)
fig = sns.heatmap(iris1.corr(),annot=True,cmap='cubehelix',linewidths=1, linecolor=
plt.show())

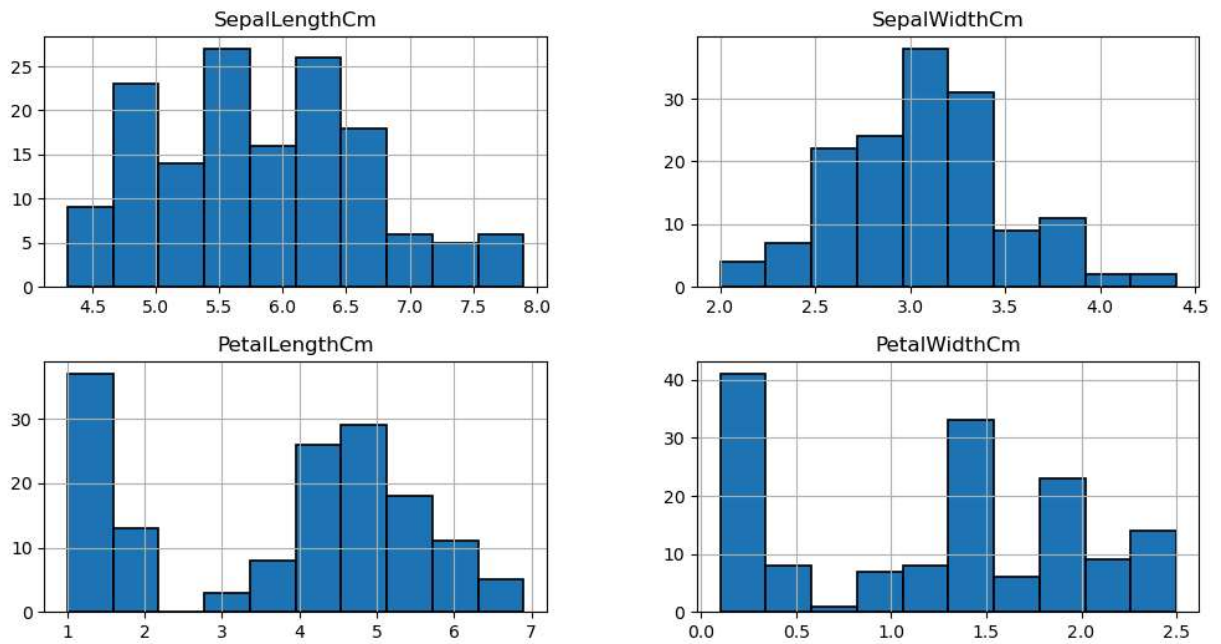
# iris.corr() by default takes numeric values only.
# but if original file has modified or due to some other reason it was giving error
# created new variable to solve problem
```



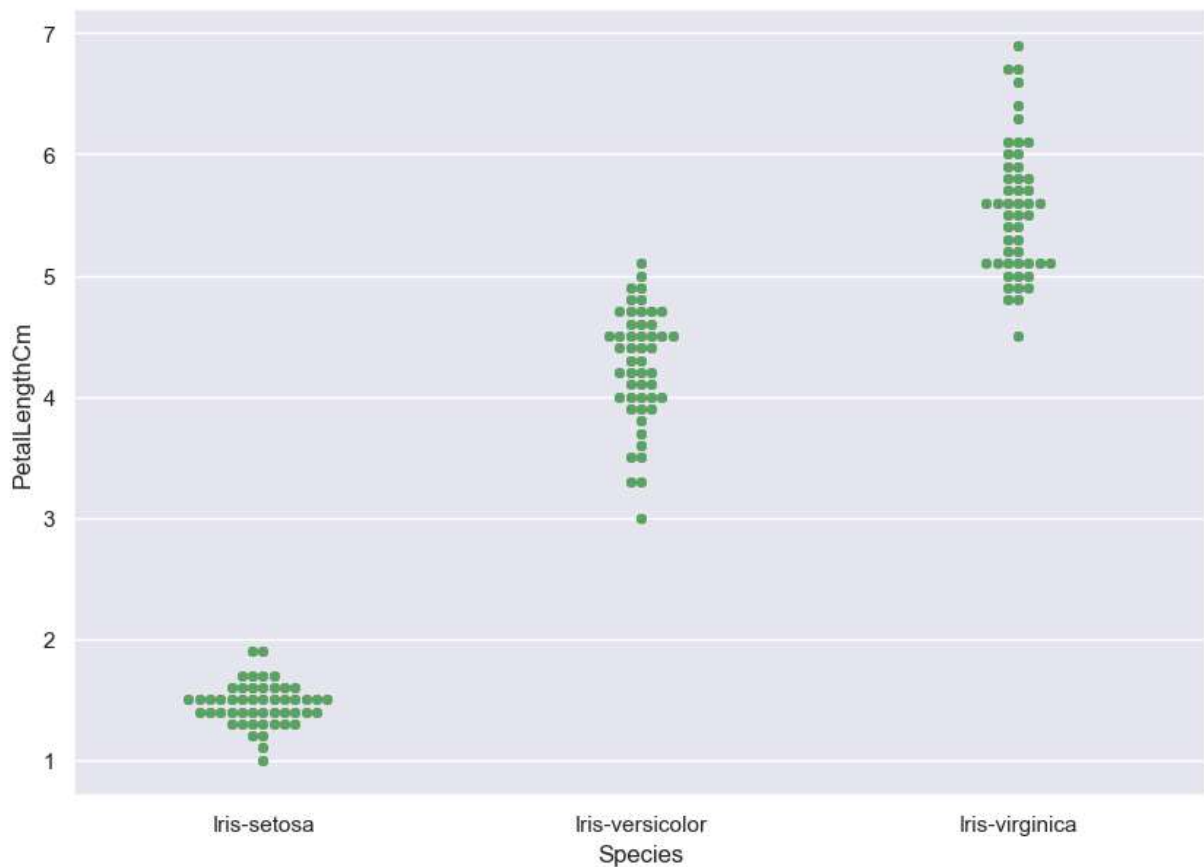
```
In [233... iris.hist(edgecolor='k',linewidth=1.2)
fig= plt.gcf()
fig.set_size_inches(12,6)

plt.show()
```



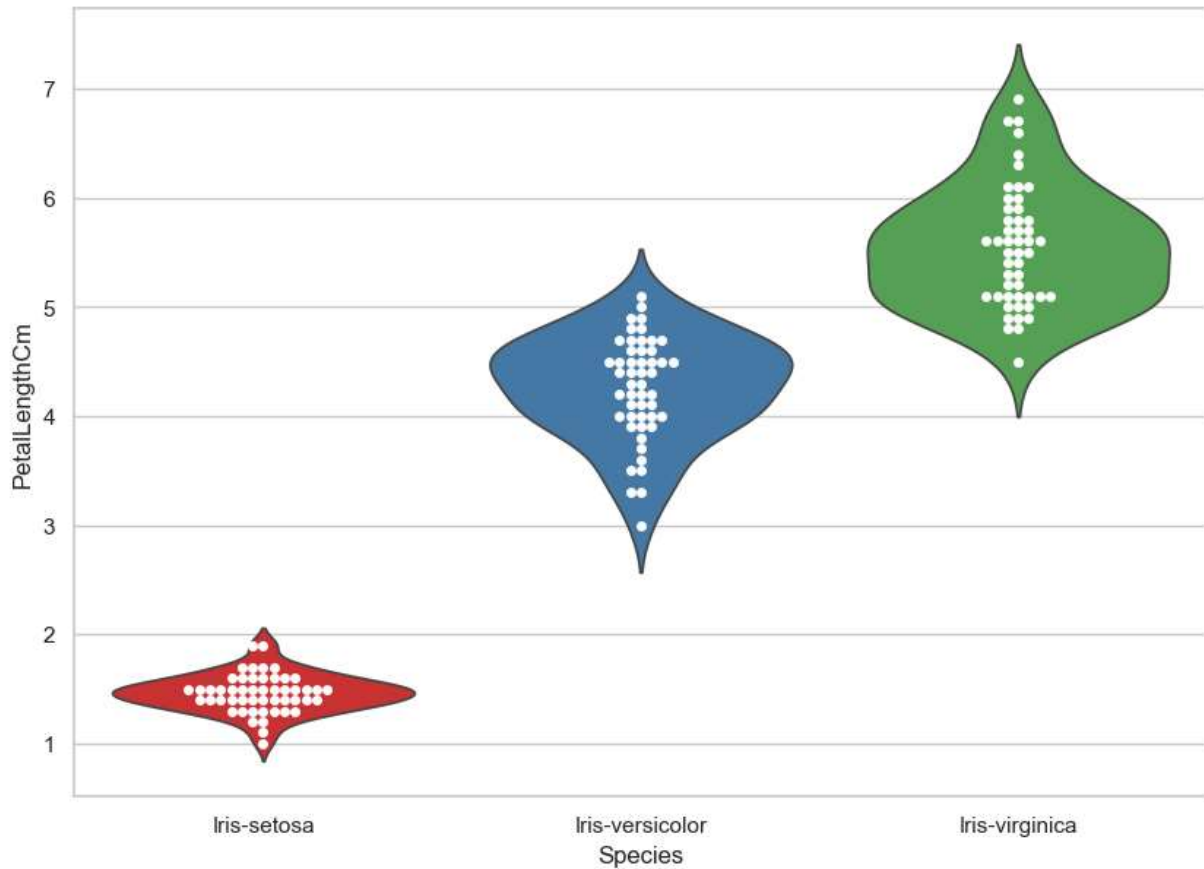


```
In [239... sns.set(style='darkgrid')
fig=plt.gcf()
fig.set_size_inches(10,7)
fig = sns.swarmplot(x='Species', y='PetalLengthCm',data=iris)
plt.show()
```



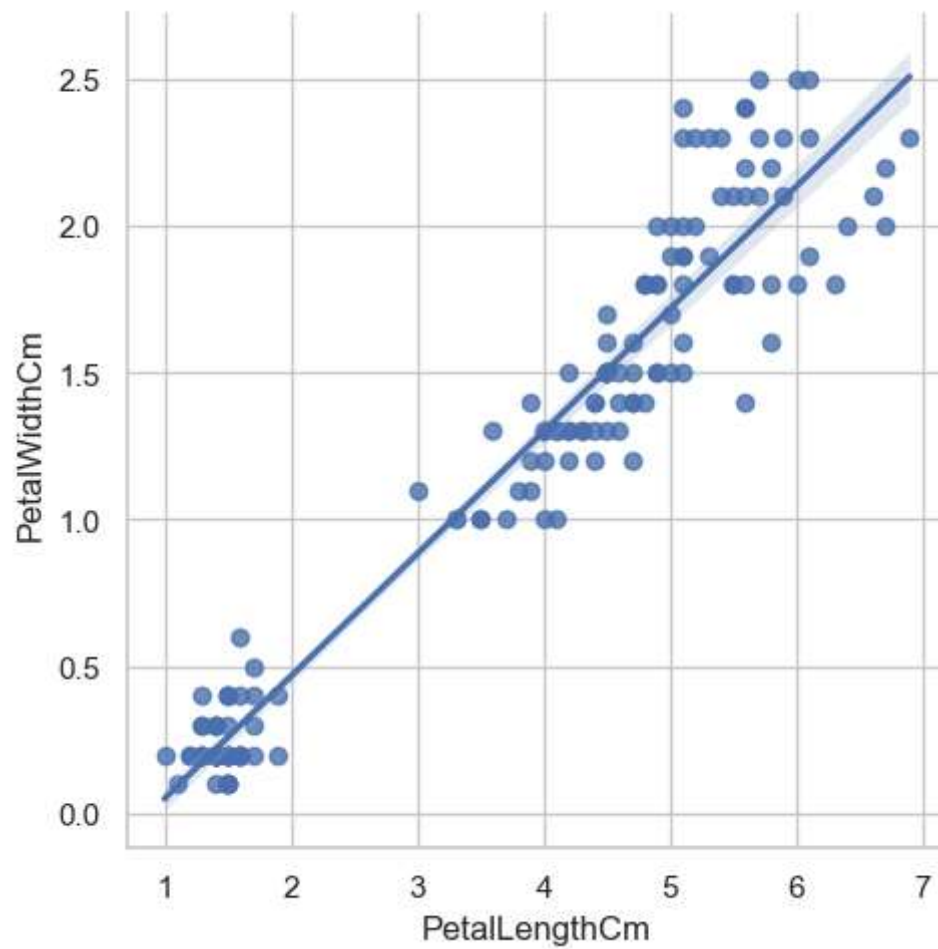
```
In [245... sns.set(style='whitegrid')
fig=plt.gcf()
fig.set_size_inches(10,7)
```

```
fig = sns.swarmplot(x='Species', y='PetalLengthCm', data=iris, color='white', edgecolor='black')  
fig = sns.violinplot(x='Species', y='PetalLengthCm', data=iris, inner=None, palette='S')  
plt.show()
```

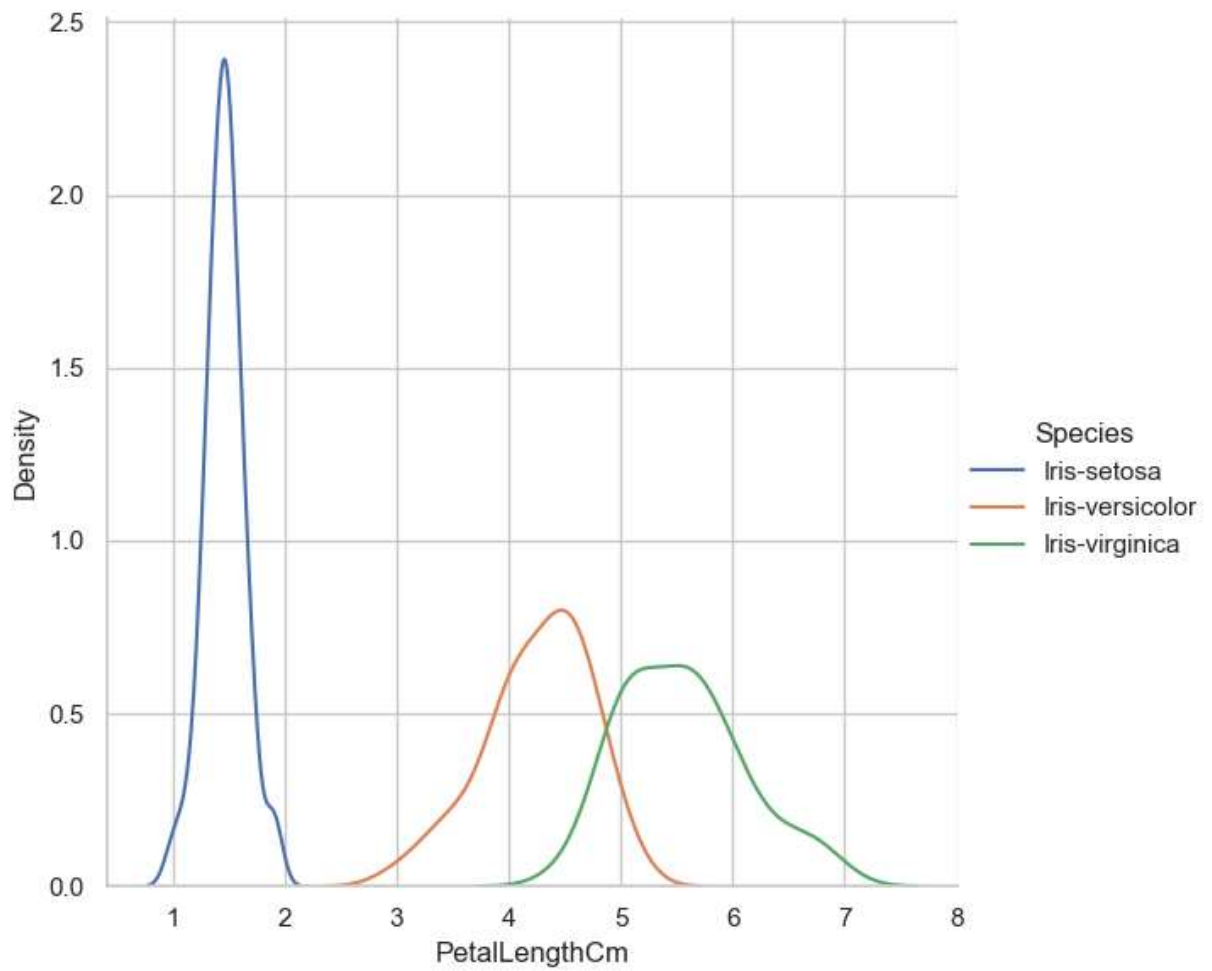


In [251...

```
fig = sns.lmplot(x="PetalLengthCm", y="PetalWidthCm", data=iris)  
plt.show()
```

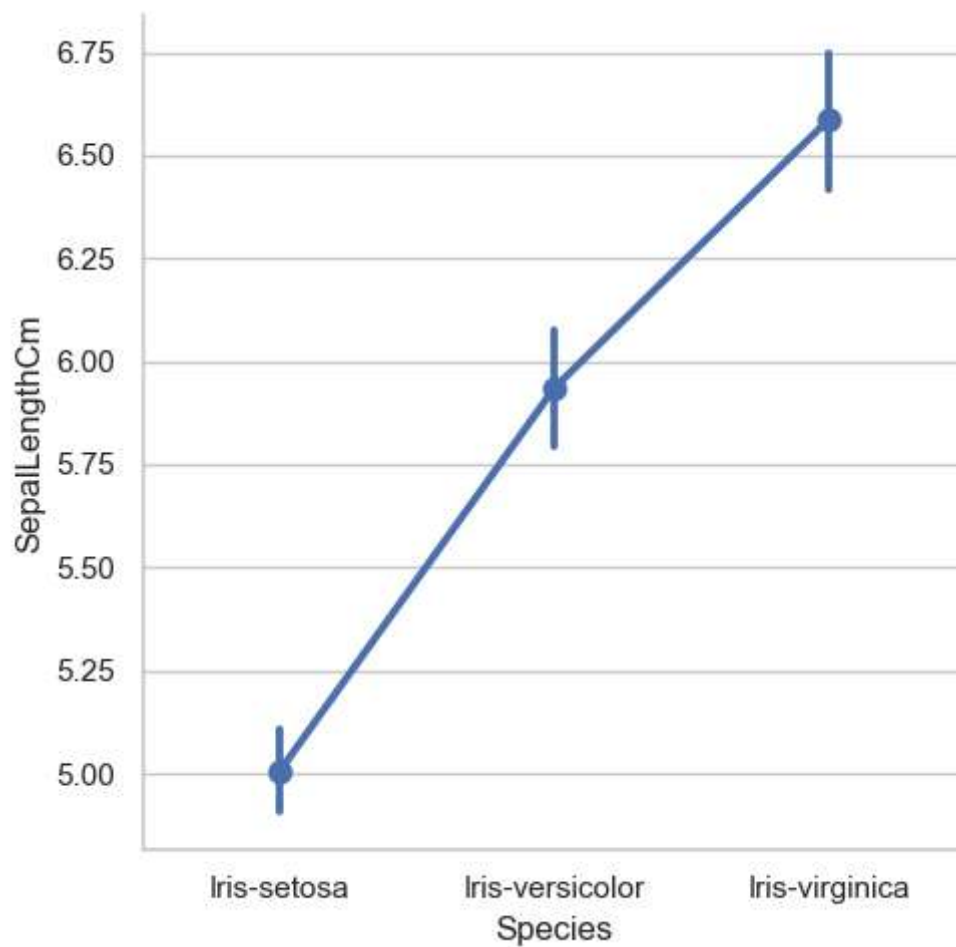


```
In [269... sns.FacetGrid(iris,hue='Species',height=6)\  
            .map(sns.kdeplot,'PetalLengthCm')\  
            .add_legend()  
            plt.show()
```

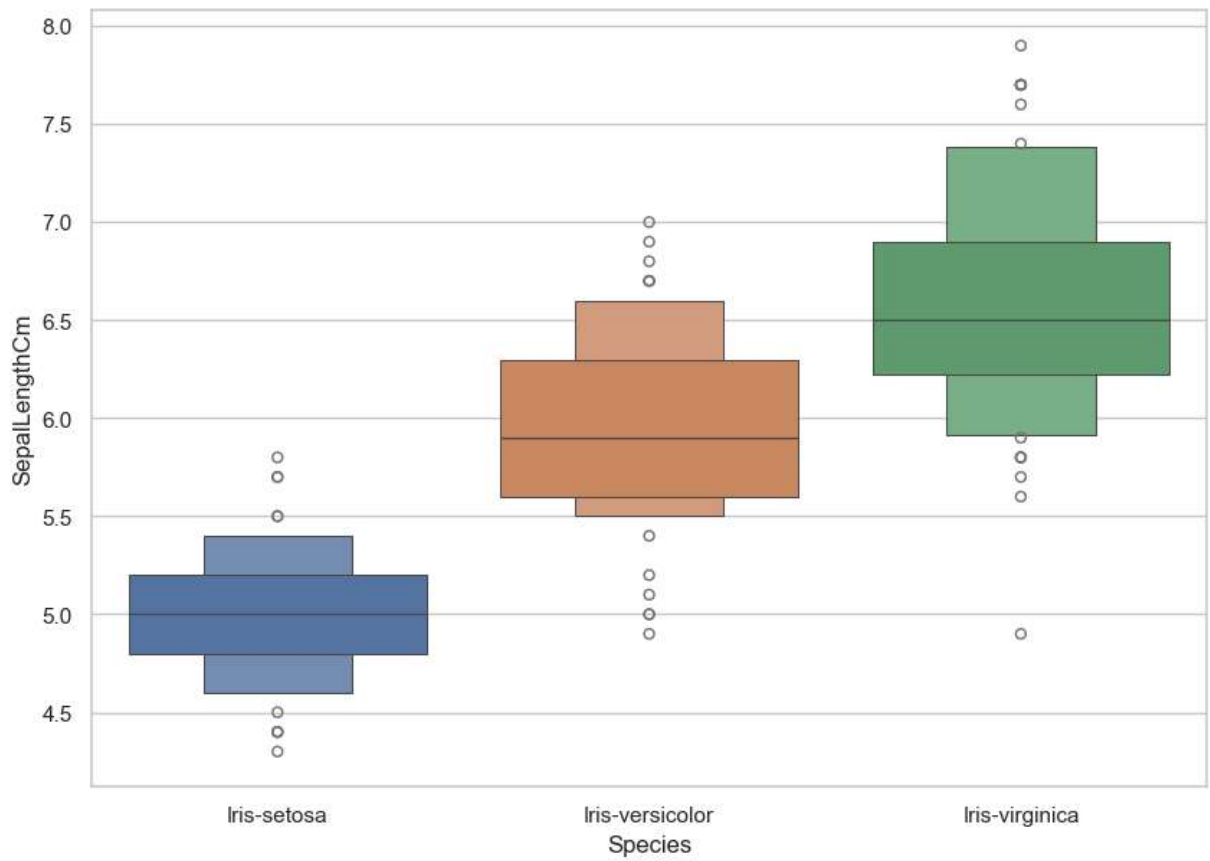


```
In [277... # sns.factorplot
# in newer version factorplot is deprecated and replaced by catplot

sns.catplot(iris,x='Species',y='SepalLengthCm',kind='point')
plt.show()
```

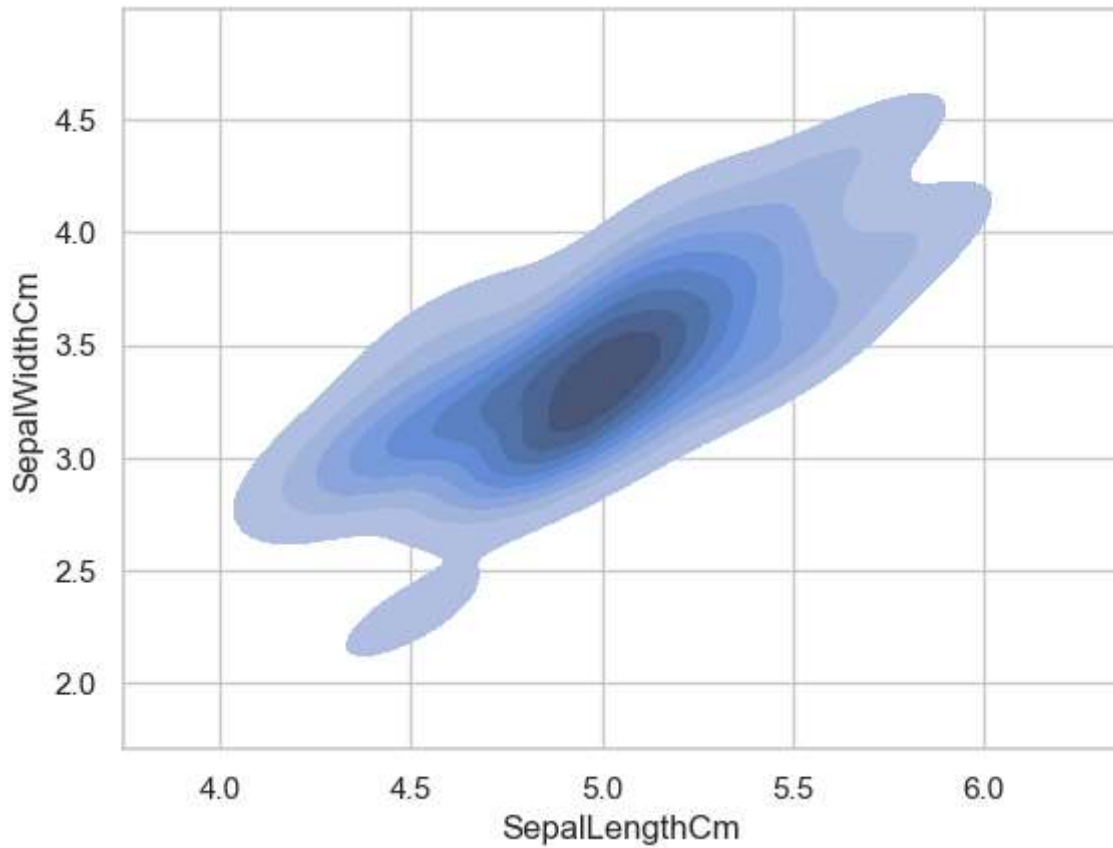


```
In [301... fig=plt.gcf()
fig.set_size_inches(10,7)
fig = sns.boxenplot(x='Species',y='SepalLengthCm',data=iris,hue='Species')
plt.show()
```



In [323...

```
# Create a kde plot of sepal_length versus sepal width for setosa species of flower  
sns.kdeplot(data=iris[iris.Species=='Iris-setosa'],x= 'SepalLengthCm',y='SepalWidthCm',  
plt.show()
```

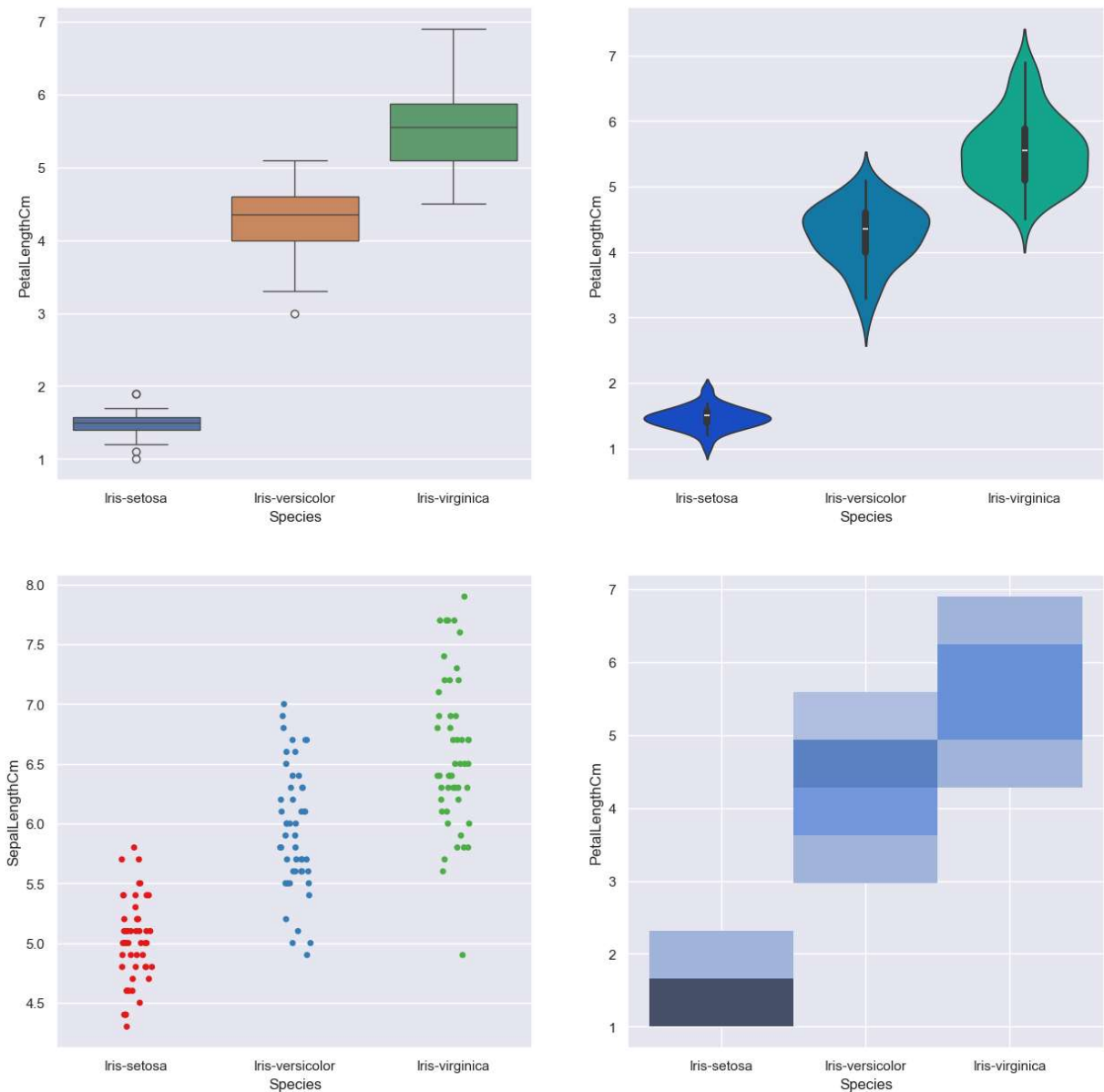



```
In [345... sns.set_style('darkgrid')
f, axes = plt.subplots(2, 2, figsize=(15, 15))

k1 = sns.boxplot(x='Species', y='PetalLengthCm', data=iris, palette='deep', ax=axes[0])
k2 = sns.violinplot(x='Species', y='PetalLengthCm', data=iris, palette='winter', ax=axes[1])
k3 = sns.stripplot(x='Species', y='SepalLengthCm', data=iris, palette='Set1', ax=axes[2])
k4 = sns.histplot(x='Species', y='PetalLengthCm', data=iris, palette='Set2', ax=axes[3])

plt.show()

# ax-argument; axes variable we created
```



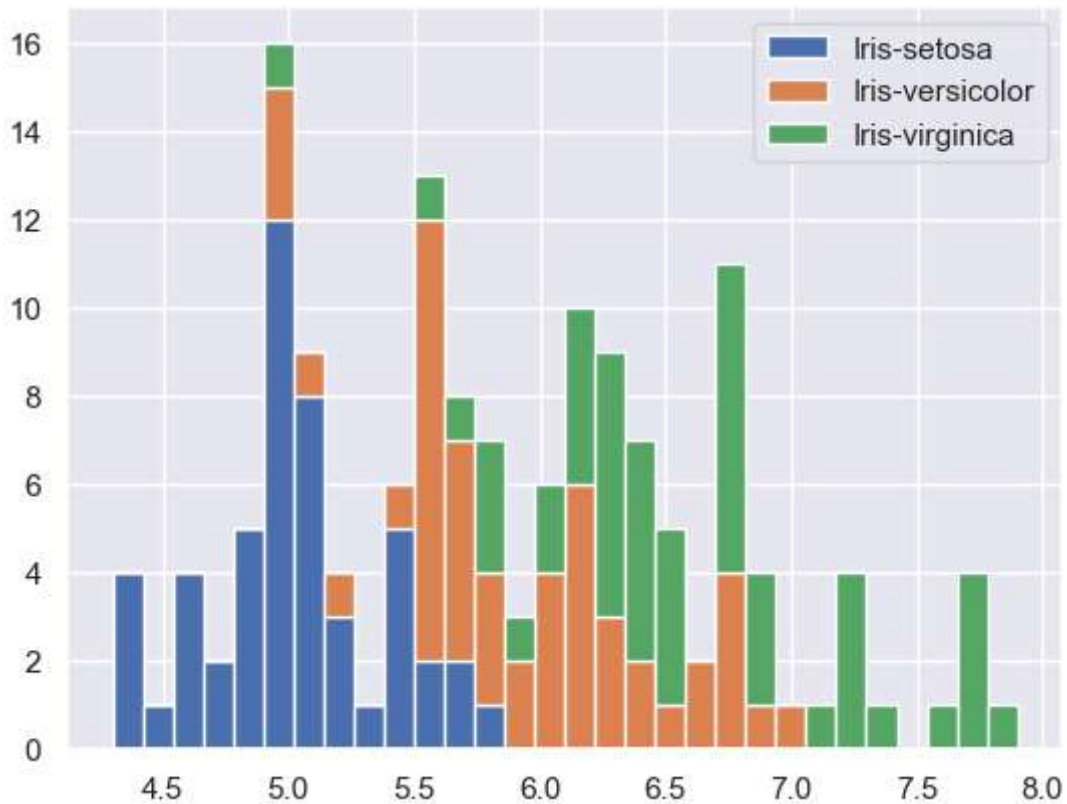
```
In [347... iris['Species']=iris['Species'].astype('category')
iris.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Sepal.LengthCm  150 non-null   float64
1   Sepal.WidthCm   150 non-null   float64
2   Petal.LengthCm  150 non-null   float64
3   Petal.WidthCm   150 non-null   float64
4   Species         150 non-null   category
dtypes: category(1), float64(4)
memory usage: 5.1 KB
```

```
In [351... list1=list()
mylabels=list()
for gen in iris.Species.cat.categories:
    list1.append(iris[iris.Species==gen].Sepal.LengthCm)    #created list for s
```

```
mylabels.append(gen) # list of species

h= plt.hist(list1, bins=30, stacked=True,rwidth=1,label=mylabels)
plt.legend()
plt.show()
```

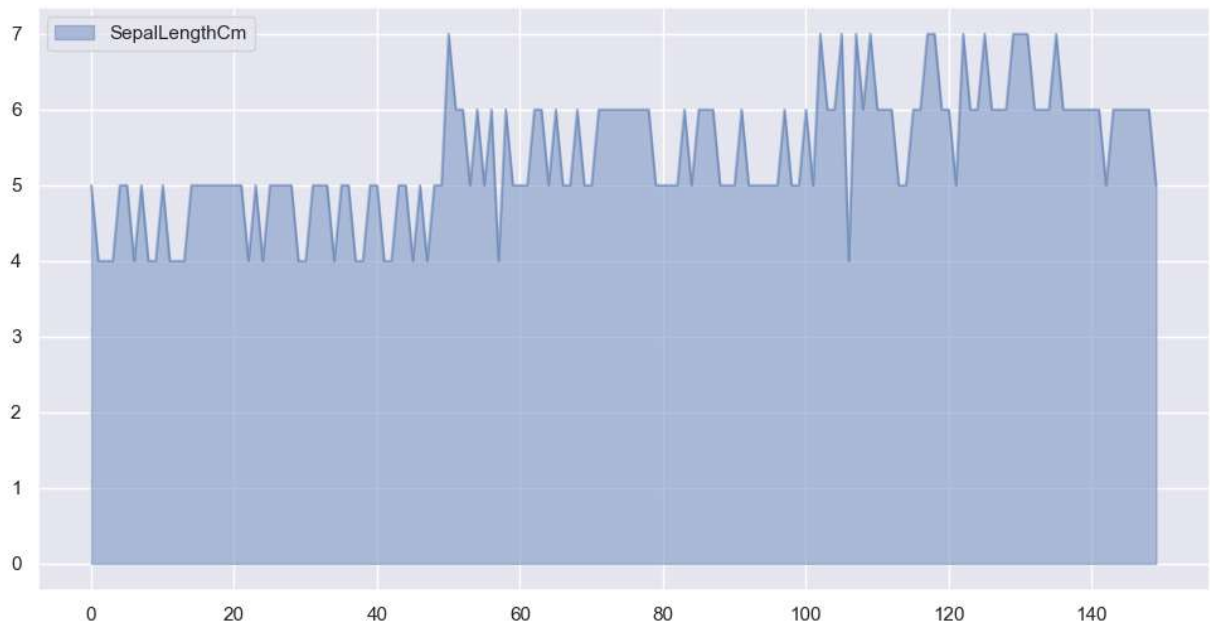


```
In [369... iris['SepalLengthCm'] = iris['SepalLengthCm'].astype('int')
iris.head()
```

```
Out[369... 
```

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	5	3.5	1.4	0.2	Iris-setosa
1	4	3.0	1.4	0.2	Iris-setosa
2	4	3.2	1.3	0.2	Iris-setosa
3	4	3.1	1.5	0.2	Iris-setosa
4	5	3.6	1.4	0.2	Iris-setosa

```
In [375... iris.plot.area(y='SepalLengthCm',alpha=0.4,figsize=(12,6))
plt.show()
```



```
In [377... iris.plot.area(y=['SepalLengthCm','SepalWidthCm','PetalLengthCm','PetalWidthCm'],al
plt.show())
```



```
In [381... sns.distplot(iris['SepalLengthCm'],kde=True,bins=20)
plt.show())
```

